SYNTHETIC DEMO - All inputs and outcomes are illustrative.

Experiment calculation memorandum

Partner growth experiment

Monitor experiment | Owner: Growth team | Protocol v1

Experiment ID: customer-business-v2-demo

Protocol saved: 2026-10-26T21:08:06+00:00 | Snapshot calculated: 2026-10-26T21:08:06+00:00

Traceable calculation memorandum. This document does not establish completed audit or approval.

Decision and experiment definition

Does a simpler offer increase revenue without increasing cancellations?

Name	Role	Definition
Control	Control	Current offer
Test A	Test	Simplified offer
Test B	Test	Simplified offer with reminder

Unit	Decision Scope	Start Date	Window Days	Maturity Delay Days
Customer	Overall	2026-10-26	28	0

Unit	Eligible Per Week	Max Recruitment Weeks
Customer	2500	12

Metric definitions and decision thresholds

revenue - Primary / Continuous / Higher is better

Name	Unit	Definition	Window Days	Effect
revenue	currency / customer	Total per randomized customer over the observation window	28	5

Name	Effect Scale	Margin	Alert Threshold	Include In Plan
revenue	Relative %	0	0	True

cancellation - Guardrail / Binary / Lower is better

Name	Unit	Definition	Window Days	Effect
cancellation	rate (0 to 1)	Cancellation within 28 days per randomized customer	28	1

Name	Effect Scale	Margin	Alert Threshold	Include In Plan
cancellation	Percentage points	0.03	0.02	True

Population and frozen weights

Fixed-weight Partner x Segment mean difference; variance = sum(w^2 s^2/n); Welch-Satterthwaite df. Binary: fixed-weight risk difference with normal inference when adequate, exact per-cell fallback.

Partner	Segment	Weight
Partner A	New	0.25
Partner A	Existing	0.25
Partner B	New	0.25
Partner B	Existing	0.25

Design assumptions

Alpha	Power	Multiplicity	Attrition	Allocation
0.05	0.8	True	0	Recommended

Alpha	Assumption Only	Historical Start	Historical End
0.05	False		

Predeclared comparison family: 4. Planning alpha per comparison: 0.0125. Power is per planned comparison.

Arm	Partner	Segment	Metric	Source
Test A	Partner A	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner A	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner A	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner A	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner B	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner B	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner B	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)

Arm	Partner	Segment	Metric	Source
Test A	Partner B	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)

Sample requirements and timing

Recruitment targets include the saved missing-outcome reserve.

Arm	Analyzable N	Recruitment N
Control	2231	2231
Test A	2230	2230
Test B	2230	2230

Recruitment duration: 2.6764 weeks. Recruitment end: 2026-11-14. Earliest mature readout: 2026-12-12. Within maximum recruitment duration: True.

Partner	Segment	Arm	Analyzable N	Recruitment N
Partner A	New	Control	558	558
Partner A	Existing	Control	558	558
Partner B	New	Control	558	558
Partner B	Existing	Control	557	557
Partner A	New	Test A	558	558
Partner A	Existing	Test A	558	558
Partner B	New	Test A	557	557
Partner B	Existing	Test A	557	557
Partner A	New	Test B	558	558
Partner A	Existing	Test B	558	558
Partner B	New	Test B	557	557
Partner B	Existing	Test B	557	557

Partner	Expected Share
Partner A	0.33333333
Partner A	0.33333333
Partner B	0.33373206
Partner B	0.33333333
Partner A	0.33333333
Partner A	0.33333333
Partner B	0.33313397
Partner B	0.33333333
Partner A	0.33333333
Partner A	0.33333333
Partner B	0.33313397
Partner B	0.33333333

Pairwise Neyman and shared-control calculation

First solve each Test versus Control using the Primary metric SD ratio and all included metric constraints. Take the largest pairwise Control N. Fix it, solve each Test again, then verify all final integer stratum counts. Business-fixed allocation instead scales its declared ratios.

Scope	Test	Pair Control N	Pair Test N	Neyman Ratio
Overall	Test A	2231	2231	1
Overall	Test B	2231	2231	1

Scope	Metric	Test	Control N	Test N
Overall	revenue	Test A	2231	2230
Overall	revenue	Test B	2231	2230
Overall	cancellation	Test A	2231	2230
Overall	cancellation	Test B	2231	2230

Scope	Detectable Change Or Margin	Planning Alpha	Power	Variance Of Difference	Degrees Of Freedom
Overall	6.125	0.0125	0.99966712	1.0765749	4304.7006
Overall	6.125	0.0125	0.99966712	1.0765749	4304.7006
Overall	0.03	0.0125	0.80005821	8.0699456e-05	4452.9865
Overall	0.03	0.0125	0.80005821	8.0699456e-05	4452.9865

Monitoring snapshot

Source: Customer records. Data updated: 2026-10-20T00:00:00+00:00. Downloading this memo does not refresh that timestamp.

Arm	Assigned N	Mature N	Observed N	Recruitment Target
Control	560	560	560	2231
Test A	520	520	520	2230
Test B	520	520	520	2230

Arm	Analyzable Target	Missing Result Rate
Control	2231	0
Test A	2230	0
Test B	2230	0

Sample ratio mismatch

No mismatch detected: Assigned N versus frozen, stratum-specific allocation; diagnostic family threshold divided across Overall and observed strata.

Scope	Status	Method	P Value	Threshold
Overall	No mismatch detected	Conditional stratified multinomial Monte Carlo (49,999 draws; seed 142857)	0.36876	0.0002
Partner A / New	No mismatch detected	Pearson goodness-of-fit	0.77880078	0.0002
Partner A / Existing	No mismatch detected	Pearson goodness-of-fit	0.77880078	0.0002
Partner B / New	No mismatch detected	Pearson goodness-of-fit	0.78817182	0.0002
Partner B / Existing	No mismatch detected	Pearson goodness-of-fit	0.77880078	0.0002

Scope	Observed	Expected
Overall	{"Control": 560, "Test A": 520, "Test B": 520}	{"Control": 533.4928229665071, "Test A": 533.2535885167463, "Test B": 533.2535885167463}
Partner A / New	{"Control": 140, "Test A": 130, "Test B": 130}	{"Control": 133.33333333333331, "Test A": 133.33333333333331, "Test B": 133.33333333333331}
Partner A / Existing	{"Control": 140, "Test A": 130, "Test B": 130}	{"Control": 133.33333333333331, "Test A": 133.33333333333331, "Test B": 133.33333333333331}
Partner B / New	{"Control": 140, "Test A": 130, "Test B": 130}	{"Control": 133.49282296650716, "Test A": 133.2535885167464, "Test B": 133.2535885167464}
Partner B / Existing	{"Control": 140, "Test A": 130, "Test B": 130}	{"Control": 133.33333333333331, "Test A": 133.33333333333331, "Test B": 133.33333333333331}

Pre-treatment balance

Review balance: SMD = (Test mean - Control mean) / sqrt((Control variance + Test variance)/2). Categorical levels are indicator proportions. The reference line is diagnostic, not proof of randomization failure. Checks are within Partner x Segment.

Partner	Segment	Feature	Category	Test
Partner A	New	baseline_age		Test A
Partner A	New	baseline_age		Test B
Partner A	Existing	baseline_age		Test A
Partner A	Existing	baseline_age		Test B
Partner B	New	baseline_age		Test A
Partner B	New	baseline_age		Test B
Partner B	Existing	baseline_age		Test A
Partner B	Existing	baseline_age		Test B
Partner A	New	baseline_channel	Branch	Test A
Partner A	New	baseline_channel	Branch	Test B
Partner A	New	baseline_channel	Online	Test A
Partner A	New	baseline_channel	Online	Test B
Partner A	Existing	baseline_channel	Branch	Test A
Partner A	Existing	baseline_channel	Branch	Test B
Partner A	Existing	baseline_channel	Online	Test A
Partner A	Existing	baseline_channel	Online	Test B
Partner B	New	baseline_channel	Branch	Test A
Partner B	New	baseline_channel	Branch	Test B

Partner	Segment	Feature	Category	Test
Partner B	New	baseline_channel	Online	Test A
Partner B	New	baseline_channel	Online	Test B
Partner B	Existing	baseline_channel	Branch	Test A
Partner B	Existing	baseline_channel	Branch	Test B
Partner B	Existing	baseline_channel	Online	Test A
Partner B	Existing	baseline_channel	Online	Test B

Partner	Control	Smd	Status	Control Mean	Test Mean
Partner A	Control	-0.12284677	Review balance	47.814286	45.930769
Partner A	Control	-0.19939898	Review balance	47.814286	44.776923
Partner A	Control	0.21497811	Review balance	44.95	48.423077
Partner A	Control	0.17840701	Review balance	44.95	47.630769
Partner B	Control	0.016584318	Within reference line	45.771429	46.023077
Partner B	Control	0.1925243	Review balance	45.771429	48.630769
Partner B	Control	-0.19286623	Review balance	50.185714	47.2
Partner B	Control	-0.18512654	Review balance	50.185714	47.246154
Partner A	Control	0.16545108	Review balance	0.47142857	0.55384615
Partner A	Control	0.041794849	Within reference line	0.47142857	0.49230769
Partner A	Control	-0.16545108	Review balance	0.52857143	0.44615385
Partner A	Control	-0.041794849	Within reference line	0.52857143	0.50769231
Partner A	Control	-0.12233951	Review balance	0.50714286	0.44615385
Partner A	Control	0.0010990222	Within reference line	0.50714286	0.50769231
Partner A	Control	0.12233951	Review balance	0.49285714	0.55384615
Partner A	Control	-0.0010990222	Within reference line	0.49285714	0.49230769
Partner B	Control	-0.057189562	Within reference line	0.52857143	0.5
Partner B	Control	-0.088004807	Within reference line	0.52857143	0.48461538
Partner B	Control	0.057189562	Within reference line	0.47142857	0.5
Partner B	Control	0.088004807	Within reference line	0.47142857	0.51538462
Partner B	Control	-0.10025094	Review balance	0.55	0.5
Partner B	Control	-0.13112858	Review balance	0.55	0.48461538
Partner B	Control	0.10025094	Review balance	0.45	0.5
Partner B	Control	0.13112858	Review balance	0.45	0.51538462

Partner	Control Missing Rate	Test Missing Rate
Partner A	0	0

Partner	Control Missing Rate	Test Missing Rate
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner A	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0
Partner B	0	0

Guardrail monitoring and stratum coverage

Scope	Metric	Test	Effect	Alert Threshold
Overall	cancellation	Test A	-0.009478022	0.02
Overall	cancellation	Test B	0.028983516	0.02

Scope	Status	Note
Overall	No point-estimate alert	Descriptive business alert; not a final safety or non-inferiority conclusion.
Overall	Operational alert	Descriptive business alert; not a final safety or non-inferiority conclusion.

Partner	Segment	Arm	Metric	Analyzable N
Partner A	New	Control	revenue	558
Partner A	New	Control	cancellation	558
Partner A	Existing	Control	revenue	558
Partner A	Existing	Control	cancellation	558
Partner B	New	Control	revenue	558
Partner B	New	Control	cancellation	558
Partner B	Existing	Control	revenue	557
Partner B	Existing	Control	cancellation	557
Partner A	New	Test A	revenue	558
Partner A	New	Test A	cancellation	558
Partner A	Existing	Test A	revenue	558
Partner A	Existing	Test A	cancellation	558
Partner B	New	Test A	revenue	557
Partner B	New	Test A	cancellation	557
Partner B	Existing	Test A	revenue	557
Partner B	Existing	Test A	cancellation	557
Partner A	New	Test B	revenue	558
Partner A	New	Test B	cancellation	558
Partner A	Existing	Test B	revenue	558
Partner A	Existing	Test B	cancellation	558
Partner B	New	Test B	revenue	557
Partner B	New	Test B	cancellation	557
Partner B	Existing	Test B	revenue	557
Partner B	Existing	Test B	cancellation	557

Partner	Observed N	Target Met
Partner A	140	False
Partner A	140	False
Partner A	140	False
Partner A	140	False
Partner B	140	False
Partner B	140	False
Partner B	140	False
Partner B	140	False
Partner A	130	False

Partner	Observed N	Target Met
Partner A	130	False
Partner A	130	False
Partner A	130	False
Partner B	130	False
Partner B	130	False
Partner B	130	False
Partner B	130	False
Partner A	130	False
Partner A	130	False
Partner A	130	False
Partner A	130	False
Partner B	130	False
Partner B	130	False
Partner B	130	False
Partner B	130	False

Assumptions, gaps and limitations

- Power uses plug-in historical moments and an approximate fixed-weight stratified Welch / normal model; historical parameter uncertainty is not fully propagated.

- Target power applies to each planned comparison, not to simultaneous success of every metric or arm.

- Pairwise Neyman with shared control is intuitive and verified, but is not a global minimum-total-sample optimization.

- Recruitment dates assume stable eligible traffic and adequate recruitment in every selected stratum.

- Fixed-horizon inference: repeated monitoring does not authorize early efficacy stopping.

- Binary planning assumes a common absolute risk difference across strata at the design alternative, or zero true difference for Guardrail non-inferiority power; rates are never silently clipped.

- The historical period was not fully supplied. Confirm the source is representative and uses the configured metric observation windows.

- Baseline covariates are assumed to be measured before assignment; timing is not independently verified.

Technical appendix: formulas and complete inputs

Tables display eight significant digits. Companion JSON retains full numeric precision for independent recalculation.

- Independent randomization units; mutually exclusive Partner x Segment strata; fixed weights identical in every arm.

- Estimate = sum_h w_h (mean_Test,h - mean_Control,h). Variance = sum_h w_h^2 (s_Test,h^2/n_Test,h + s_Control,h^2/n_Control,h).

- Welch-Satterthwaite df = variance^2 / sum_h,arm [(w_h^2 s_h,arm^2/n_h,arm)^2/(n_h,arm-1)].

- Continuous planning: noncentral t approximation with the same integer cell counts and df. Binary planning and large-cell testing: fixed-weight risk difference normal approximation; sparse cells require exact procedures.

- Continuous Guardrail: benefit-oriented difference compared with -margin; non-inferiority p-value is one-sided. Decisions use conservative family intervals, not p > 0.05.

- Adequate Binary cells use fixed-weight normal risk-difference intervals with family protection. Sparse cells use Newcombe-Wilson cell intervals with additional Bonferroni protection over strata; Fisher p-values do not invert these intervals.

- SMD uses pre-treatment covariates only. SRM uses assigned counts and frozen stratum allocation. Missing checks are never passes.

- Historic sample variance uses ddof=1 / VAR.S. Missing outcomes are not zeros; observed outcomes are not assignment counts.

Partner	Segment	Metric	Arm	Sample Mean
Partner A	New	revenue	Control	100
Partner A	New	cancellation	Control	0.1
Partner A	Existing	revenue	Control	120
Partner A	Existing	cancellation	Control	0.1
Partner B	New	revenue	Control	125
Partner B	New	cancellation	Control	0.1
Partner B	Existing	revenue	Control	145
Partner B	Existing	cancellation	Control	0.1

Partner	Sample Variance	Sample Number	Successes
Partner A	900	1200	Not available
Partner A	0.090075063	1200	120
Partner A	1156	1200	Not available
Partner A	0.090075063	1200	120
Partner B	1225	1200	Not available
Partner B	0.090075063	1200	120
Partner B	1521	1200	Not available
Partner B	0.090075063	1200	120

Partner	Segment	Metric	Arm	Sample Mean
Partner A	New	revenue	Control	95.739214
Partner A	New	cancellation	Control	0.12142857
Partner A	New	revenue	Test A	105.73438
Partner A	New	cancellation	Test A	0.092307692
Partner A	New	revenue	Test B	106.51638
Partner A	New	cancellation	Test B	0.16153846
Partner A	Existing	revenue	Control	120.76621

Partner	Segment	Metric	Arm	Sample Mean
Partner A	Existing	cancellation	Control	0.10714286
Partner A	Existing	revenue	Test A	123.79754
Partner A	Existing	cancellation	Test A	0.092307692
Partner A	Existing	revenue	Test B	130.83285
Partner A	Existing	cancellation	Test B	0.11538462
Partner B	New	revenue	Control	123.93043
Partner B	New	cancellation	Control	0.092857143
Partner B	New	revenue	Test A	130.13515
Partner B	New	cancellation	Test A	0.076923077
Partner B	New	revenue	Test B	135.44292
Partner B	New	cancellation	Test B	0.14615385
Partner B	Existing	revenue	Control	151.29757
Partner B	Existing	cancellation	Control	0.085714286
Partner B	Existing	revenue	Test A	149.226
Partner B	Existing	cancellation	Test A	0.10769231
Partner B	Existing	revenue	Test B	153.245
Partner B	Existing	cancellation	Test B	0.1

Partner	Sample Variance	Sample Number	Successes
Partner A	947.07903	140	Not available
Partner A	0.10745118	140	17
Partner A	914.0532	130	Not available
Partner A	0.084436494	130	12
Partner A	882.88135	130	Not available
Partner A	0.13649374	130	21
Partner A	1426.8227	140	Not available
Partner A	0.09635149	140	15
Partner A	1027.144	130	Not available
Partner A	0.084436494	130	12
Partner A	1003.8628	130	Not available
Partner A	0.10286225	130	15
Partner B	1040.5981	140	Not available
Partner B	0.084840699	140	13
Partner B	1119.9928	130	Not available
Partner B	0.071556351	130	10

Partner	Sample Variance	Sample Number	Successes
Partner B	1165.9504	130	Not available
Partner B	0.12576029	130	19
Partner B	1434.9173	140	Not available
Partner B	0.078931141	140	12
Partner B	1353.2164	130	Not available
Partner B	0.096839595	130	14
Partner B	1505.8474	130	Not available
Partner B	0.090697674	130	13

Reproduction fingerprints

Python	Numpy	Pandas	Scipy	Statsmodels
3.11.15	1.26.4	2.2.2	1.13.1	0.14.2

Python	Streamlit	Reportlab
3.11.15	1.39.0	4.4.9

Code: customer-v2-sha256:22a48312d3977f1fa7ce93127e4d1d4909fe104bb255b0eb94104a9d03f2cf0a
Configuration: b224399aed519178fb3e87d8c66c355911e429557767faed2058115bba182b3d
Historical inputs: 6db1f0a135c8fb72cc56d20247cd1193da534f518d18d0d12c13555e438dbaec
Snapshot inputs: 4a2eea080b17ffab8ce71553d27a1e9415d93ba4424e9776ffb0777b382bd3d4

Download the accompanying protocol JSON and aggregate CSV for machine-readable inputs. Individual customer IDs are excluded.